

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	xxxxxx
Project Title	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To automate and enhance the technical interview preparation process by implementing speech-to-text and AI analysis to ensure fast and accurate evaluation of spoken technical concepts.
Scope	The project comprehensively assesses a student's technical articulation, utilizing Gemini AI to catch concept gaps and track spoken fluency metrics.
Problem Statement	
Description	Manual evaluation of verbal communication and technical articulation is highly inefficient and impossible to scale for hundreds of students weekly.
Impact	Unaddressed communication gaps cause technically sound students to lose job placement opportunities, negatively impacting college placement metrics.
Proposed Solution	
Approach	Employing generative AI and speech processing to automatically transcribe, evaluate, and score spoken concept explanations.
Key Features	-Instant detection of missing technical concepts and nervous filler words ("um", "uh").

	- AI-powered automatic speech-to-text transcription. - Real-time analytics dashboard with downloadable feedback reports.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU (for processing speech models)
Memory	RAM specifications	8 GB/16 GB
Storage	Disk space for data, models, and logs	512 GB SSD/1 TB SSD
Software		
Frameworks	Python frameworks	Streamlit / FastAPI
Libraries	Core python libraries	SpeechRecognition, pydantic, requests
Development Environment	IDE	VS Code / Google Colab
Data		
Data	Source, size, format	OpenAI Whisper API (Audio-to-Text), Gemini AI API (Evaluation text parsing)